
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Naggenda, Geoffrey
Date of birth:	9/23/1990
Subject:	020-AGK
Result:	40 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2876		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
2	2739		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	3651		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	2888		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
5	2754		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
6	2907		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
7	2769		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
8	3639		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
9	741	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
10	2420	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
11	2414	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
12	900	21.3.2.2	System components: etc	0,00	1,00
13	1175	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
14	1323	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	1,00	1,00
15	2919	21.8	FUEL SYSTEM	0,00	1,00
16	1074	21.8.2.2	Design, operation, system components, etc	1,00	1,00
17	913	21.9.1.7	Circuit breakers	0,00	1,00
18	159	21.9.3	Generation	0,00	1,00
19	1164	21.11.2.4	Turbine	1,00	1,00
20	1091	21.13	OXYGEN SYSTEMS	0,00	1,00
21	34	22.	INSTRUMENTATION	1,00	1,00
22	3871	22.	INSTRUMENTATION	1,00	1,00
23	4464	22.	INSTRUMENTATION	0,00	1,00
24	3094	22.	INSTRUMENTATION	0,00	1,00
25	3014	22.	INSTRUMENTATION	0,00	1,00
26	3100	22.	INSTRUMENTATION	0,00	1,00
27	3405	22.	INSTRUMENTATION	0,00	1,00
28	2822	22.	INSTRUMENTATION	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	2793	22.	INSTRUMENTATION	0,00	1,00
30	3112	22.	INSTRUMENTATION	0,00	1,00
31	2819	22.	INSTRUMENTATION	1,00	1,00
32	4028	22.	INSTRUMENTATION	0,00	1,00
33	1123	22.2	MEASUREMENT OF AIR DATA PARAMETERS	1,00	1,00
34	764	22.2.1	Pressure measurement	1,00	1,00
35	694	22.2.4	Altimeter	0,00	1,00
36	1024	22.2.4	Altimeter	0,00	1,00
37	107	22.2.6	Airspeed indicator (ASI)	1,00	1,00
38	1690	22.2.6	Airspeed indicator (ASI)	1,00	1,00
39	87	22.2.6	Airspeed indicator (ASI)	1,00	1,00
40	1711	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
Total:40%				16,00	40,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Naggenda, Geoffrey

Location: Uganda Civil Aviation Authority

Your Result: 16,00 from 40,00 Points (40,00 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	8,00	20,00	40,00
21.2. AIRFRAME	3	1,00	3,00	33,33
21.3. HYDRAULICS	1	0,00	1,00	0,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	2	1,00	2,00	50,00
21.8. FUEL SYSTEM	2	1,00	2,00	50,00
21.9. ELECTRICS	2	0,00	2,00	0,00
21.11. TURBINE ENGINES	1	1,00	1,00	100,00
21.13. OXYGEN SYSTEMS	1	0,00	1,00	0,00
22. INSTRUMENTATION	20	8,00	20,00	40,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	7	5,00	7,00	71,43
22.4. GYROSCOPIC INSTRUMENTS	1	0,00	1,00	0,00
Total	40	16,00	40,00	40,00

1

What should help prevent aircraft induced oscillation on a gyroplane?

- ☐ Adding a horizontal stabilizer.
- ☒ Lowering the center of gravity below the thrust line.
- ☐ Increasing cyclic control sensitivity.

2

Detonation in an aircraft engine is most likely to occur whenever the

- ☒ engine is operated under conditions which cause the fuel mixture to burn instantaneously.
- ☐ fuel/air ratio is such that the mixture burns extremely slow.
- ☒ fuel being used is of a higher grade than recommended by the engine manufacturer.

3

The purpose of a kill switch is to

- ☐ shut off the fuel to the carburetor.
- ☒ ground the battery eliminating current for the ignition system.
- ☐ ground the lead wire to the ignition coil shutting down the powerplant.

4

A high-frequency vibration that suddenly occurs during flight could be an indication of a defective

☐ transmission.

☒ freewheeling unit.

☐ ignition system.

5

The presence of carburetor ice in an aircraft equipped with a fixed-pitch propeller can be verified by applying carburetor heat and noting

- ☐ an increase in RPM and then a gradual decrease in RPM.
- ☒ a decrease in RPM and then a gradual increase in RPM.
- ☐ a decrease in RPM and then a constant RPM indication.

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6

What type of oxygen system is most commonly found in general aviation aircraft?

☐ Demand.

☐ Pressure demand.

☒ Continuous flow.

7

Running a fuel tank dry before switching tanks is not a good practice because

- ☒ the engine-driven fuel pump or electric fuel boost pump draw air into the fuel system and cause vapor lock.
- ☐ the engine-driven fuel pump is lubricated by fuel and operating on a dry tank may cause pump failure.
- ☐ any foreign matter in the tank will be pumped into the fuel system.

8

Fuel enters a two-cycle engine through an

☐ intake valve and reed valve.

☐ intake port and reed valve.

☒ intake port and intake valve.

9

On an aeroplane, spoilers are :

- ☐ lower wing surface devices, their deflection is always asymmetrical.
- ☐ lower wing surface devices, their deflection is symmetrical or asymmetrical.
- ☐ upper wing surface devices, their deflection is always asymmetrical.
- ☒ upper wing surface devices, their deflection is symmetrical or asymmetrical.

10

How can the pilot increase the rate of turn and decrease the radius at the same time?

- ☐ Steepen the bank and decrease airspeed.
- ☒ Steepen the bank and increase airspeed.
- ☐ Shallow the bank and increase airspeed.

11

What effect does an increase in airspeed have on a coordinated turn while maintaining a constant angle of bank and altitude?

- ☒ The rate of turn will decrease resulting in a decreased load factor.
- ☐ The rate of turn will decrease resulting in no changes in load factor.
- ☐ The rate of turn will increase resulting in an increased load factor.

12

The hydraulic oil, entering the hydraulic pump, is slightly pressurised to :

- ☒ prevent cavitation in the pump
- ☐ prevent vapour locking.
- ☐ prevent overheating of the pump.
- ☒ ensure sufficient pump output

13

The type of brake unit found on most transport aeroplanes is a:

- ☒ Drum type brake.
- ☐ Belt brake.
- ☐ Multiple disk brake.
- ☐ Single disk brake.

14

A tubeless tyre is a tyre: 1. which requires solid or branched wheels 2. whose valve can be sheared in sudden accelerations 3. whose mounting rim must be flawless 4. which requires no rim protection between rim flange and tire removing device 5. which does not burst in the event of a tire puncture 6. which eliminates internal friction between the tube and the tire The combination regrouping all the correct statements is :

☐ 3, 4, 5.

☐ 2, 3, 6.

☒ 1, 5, 6.

☐ 1, 2, 5.

15

The purpose of the preheating coil used in hot air balloons is to

- ☒ prevent ice from forming in the fuel lines.
- ☐ vaporize the fuel for more efficient burner operation.
- ☐ warm the fuel tanks for more efficient fuel flow.

16

The main purpose of the mixture control is to:

- ☐ increase the oxygen supplied to the engine
- ☒ adjust the fuel flow to obtain the proper fuel/air ratio
- ☐ decrease oxygen supplied to the engine
- ☐ decrease the air supplied to the engine

17

The function of the Generator Breaker is to close when the voltage of the :

- ☐ battery is greater than the alternator voltage and to open
- ☐ battery is greater than the generator voltage and to open
- ☒ alternator is greater than the battery voltage and to open
- ☐ generator is greater than battery voltage

18

The output of a generator is controlled by :

- ☒ varying the field strength.
- ☐ the reverse current relay circuit breaker.
- ☒ varying the length of wire in the armature windings.
- ☐ varying the speed of the engine.

19

The control of free turbine engines on turboprops, is accomplished by: - a propeller control lever used to select: 1 - propeller RPM 2 - turbine temperature 3 - turbine RPM - a fuel control lever used to select: 4 - propeller RPM 5 - torque 6 - turbine temperature
The combination which regroups all of the correct statements is :

☐ 1 - 3 - 5

☐ 2 - 4 - 5

☒ 1 - 5 - 6

☐ 3 - 4 - 6

20

The purpose of a diluter demand regulator in an oxygen system is to :

- ☐ deliver oxygen flow when inhaling.
- ☒ is only recommended with smoke in the cockpit.
- ☐ mix air and oxygen in a passenger oxygen mask.
- ☐ deliver oxygen flow only above FL 100.

21

If the Pitot head is blocked, what airspeed indication can he expected?

- ☒ **No change of IAS in level flight, even with large power changes.**
- ☐ **Constant IAS during a descent.**
- ☐ **A decrease of IAS during a climb.**

22

When an altimeter is changed from 30.11" Hg to 29.96" Hg, in which direction will the indicated altitude change and by what value?

☐ Altimeter will indicate 150 feet higher.

☒ Altimeter will indicate 150 feet lower.

☐ Altimeter will indicate 15 feet lower.

23

(Refer to Figures 139 and 141.) Which displacement from the localizer centerline and glide slope at the 1,300-foot point from the runway is indicated?

- ☒ **21 feet below the glide slope and approximately 320 feet to the right of the runway centerline.**
- ☐ **21 feet above the glide slope and approximately 320 feet to the left of the runway centerline.**
- ☐ **28 feet above the glide slope and approximately 250 feet to the left of the runway centerline.**

See Attachments:

- | | | |
|-----------------------|--------------------|-------------------|
| 1. instrument_139.jpg | instrument_139.jpg | (Attachment No.1) |
| 2. instrument_140.jpg | instrument_140.jpg | (Attachment No.2) |

24

How can a pilot identify a military airport at night?

- ☒ White and red beacon light with dual flash of the white.
- ☐ Green, yellow, and white beacon light.
- ☐ Green and white beacon light with dual flash of the white.

25

What is the significance of the blue radial line on the airspeed indicator of a light multiengine airplane and when is it to be used? It indicates the

- ☐ speed which will provide the maximum altitude gain in a given time when one engine is inoperative and should be used for climb and final approach during engine-out operations.
- ☒ speed which will provide the greatest height for a given distance of forward travel when one engine is inoperative and should be used for all climbs during engine-out operations.
- ☐ minimum speed at which the airplane is controllable when the critical engine is suddenly made inoperative and should be used at all altitudes when an engine is inoperative.

26

Identify the runway distance remaining markers.

- ☐ Signs with increments of 1,000 feet distance remaining.
- ☒ Yellow marker laterally placed across the runway with signs on the side denoting distance to end.
- ☐ Red markers laterally placed across the runway at 3,000 feet from the end.

27

How will the airspeed indicator react if the ram air input to the pitot head is blocked by ice, but the drain hole and static port are not?

- ☐ Indication will drop to zero.
- ☒ Indication will remain constant but will increase in a climb.
- ☐ Indication will rise to the top of the scale.

28

A possible result of using the emergency alternate source of static pressure inside the cabin of an unpressurized airplane is the

- ☒ altimeter may indicate an altitude lower than the actual altitude being flown.
- ☐ altimeter may indicate an altitude higher than the actual altitude being flown.
- ☐ airspeed indicator may indicate less than normal.

29

In the Northern Hemisphere, if an aircraft is accelerated or decelerated, the magnetic compass will normally indicate

- ☒ correctly when on a north or south heading while either accelerating or decelerating.
- ☐ a turn toward the south while accelerating on a west heading.
- ☒ a turn momentarily, with changes in airspeed on any heading.

30

What is the advantage of a three-bar VASI?

- ☐ A normal glide angle is afforded both high and low cockpit aircraft.
- ☒ The three-bar VASI is much more visible and can be used at a greater height.
- ☐ Pilots have a choice of glide angles.

31

If the static pressure tubes are broken inside a pressurized cabin during a high-altitude flight, the altimeter would probably indicate

☒ lower than actual flight altitude.

☐ sea level.

☐ higher than actual flight altitude.

32

When making an airborne VOR check, what is the maximum allowable tolerance between the two indicators of a dual VOR system (units independent of each other except the antenna)?

- ☐ 4 degrees between the two indicated bearings of a VOR.
- ☐ 6 degrees between the two indicated radials of a VOR.
- ☒ Plus or minus 4 degrees when set to identical radials of a VOR.

33

A VSI may suffer from error due to:

- ☐ pitot blockage, static blockage, aircraft manoeuvre
- ☒ time lag, instrument calibration, blockage
- ☐ density, aircraft manoeuvre, position of the static vent
- ☐ compressibility, time lag, blockage

34

The reported QNH of a given station is the:

- ☐ actual barometric pressure measured at sea level;
- ☒ Station's barometric pressure corrected to mean sea level pressure.
- ☐ actual barometric pressure measured at the station;

35

The high Altitude Radio Altimeter uses the following wavelengths:

☒ **hectometric.**

☐ **decimetric.**

☐ **metric.**

☐ **myriametric.**

36

QNH is the sub-scale setting which will cause the altimeter to read:

- ☐ vertical displacement above mean sea level
- ☒ the altitude at which any given pressure exists in the standard atmosphere
- ☐ pressure altitude
- ☐ vertical displacement above the airfield pressure datum

37

An aircraft is maintaining FL 120 in cloud. The ASI reading falls to zero. The most probable cause is:

☐ Pitot head and static vent blocked by ice.

☒ ASI malfunction.

☐ Static vent blocked by ice.

38

VFE is the maximum speed:

- ☐ *with the flaps extended in the take-off configuration.*
- ☐ *with the flaps extended in the landing configuration.*
- ☒ *with the flaps extended.*
- ☐ *at which the flaps can be operated when flying in turbulences.*

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39

Rectified Airspeed is:

- ☐ IAS corrected for density error.
- ☐ IAS corrected for density and compressibility errors.
- ☒ IAS corrected for instrument and pressure error.

40

During a stabilised climbing turn at a constant rate, the instruments which indicate the correct pitch and bank are the:

- ☐ altimeter and turn-and-slip indicator;
- ☒ vertical-speed indicator and turn-and-slip indicator;
- ☐ attitude indicator and turn-and-slip indicator.

Attachment No. 1

Questions 23

instrument_139.jpg

instrument_139.jpg

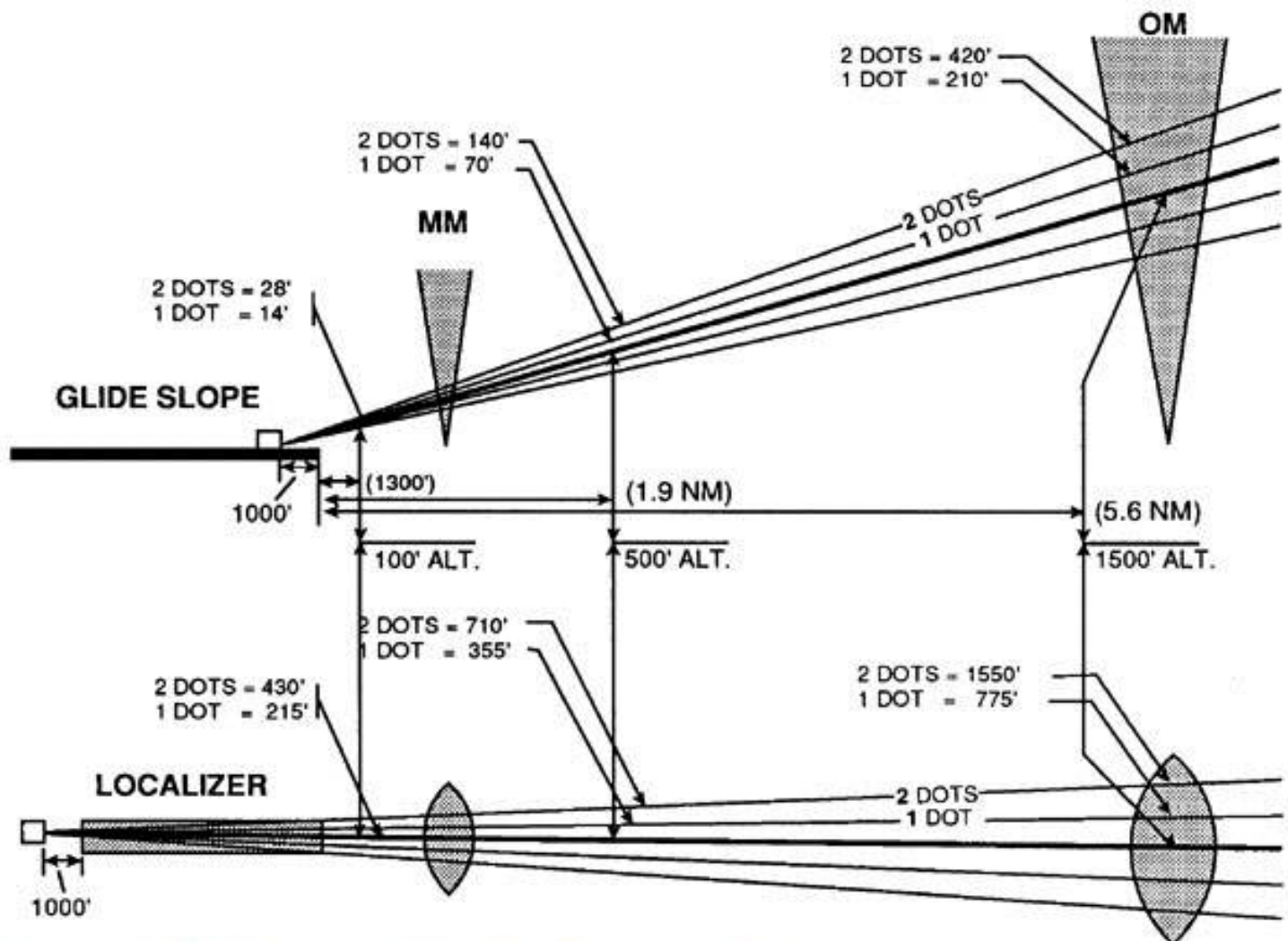


Figure 139. Glide Slope and Localizer Illustration © ASA

Attachment No. 2

Questions 23

instrument_140.jpg

instrument_140.jpg

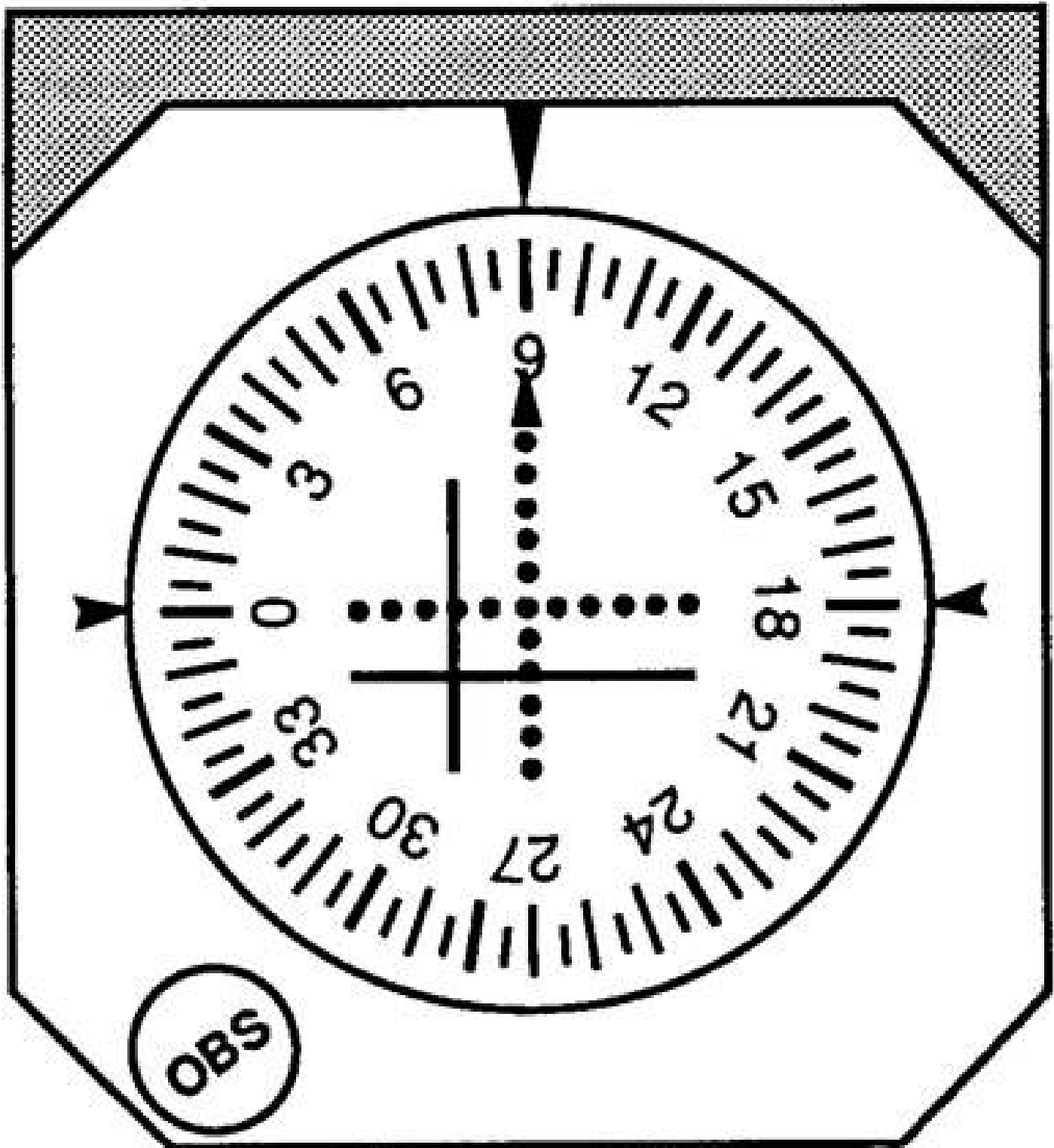


Figure 140. OBS, ILS, and GS Displacement © ASA